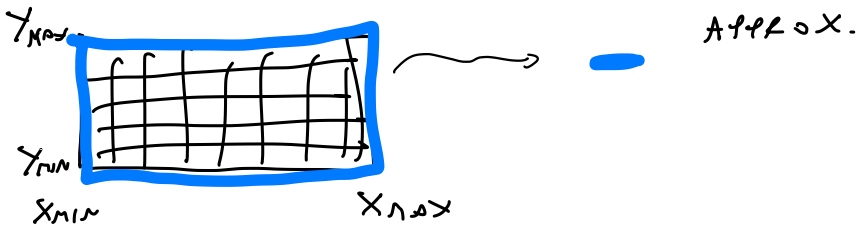
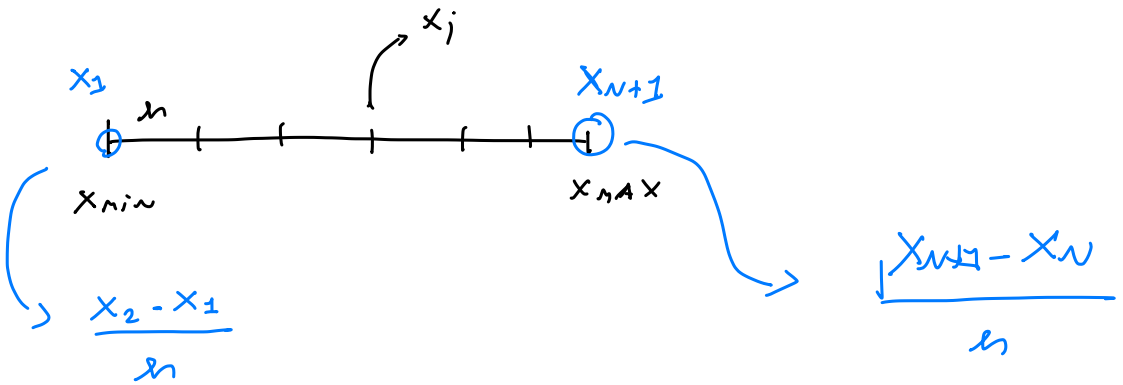
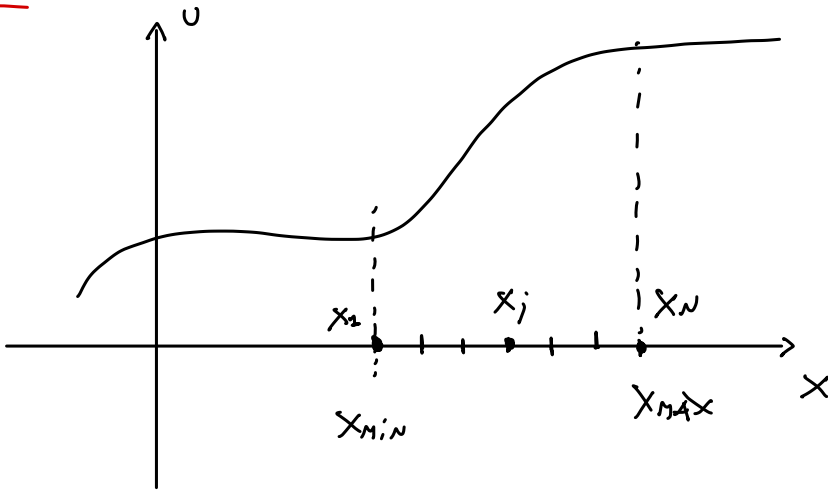


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$$\frac{1}{r} \frac{\delta \phi}{\delta r} + \frac{1}{r^2} \frac{\delta^2 \phi}{\delta \theta^2} + \frac{\delta^2 \phi}{\delta r^2} = 0$$

$$\phi_{i,j} = \phi(r_i, \theta_j)$$

$$r_i = (i-1) \Delta r \quad i = 1 \dots (N+1) \quad \Delta r = R/N \quad \phi \rightarrow \phi_{i,j}$$

$$\theta_j = (j-1) \Delta \theta \quad j = 1 \dots (M+1) \quad \Delta \theta = \pi/M$$

$$\begin{aligned} \frac{1}{r_i} \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta r} + \frac{1}{(r_i)^2} \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta \theta^2} + \\ + \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} = 0 \end{aligned}$$

discretized PDE $\rightarrow$ - valid in the circular domain

- NOT valid at the boundaries

on $\phi_{N+1,j} = \phi_{1,j}$

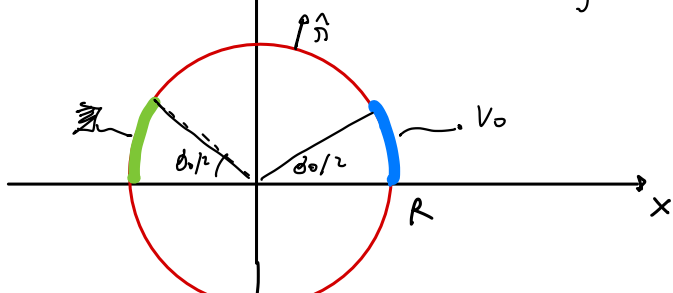
$$\frac{\phi_{N+1,j} - \phi_{1,j}}{\Delta r} = 0$$

$$\bar{\mathbf{j}} \cdot \hat{\mathbf{n}} \, ds = 0$$

$$\bar{\mathbf{j}} \cdot \hat{\mathbf{n}} = 0 \quad \hat{\mathbf{n}} = \hat{\mathbf{n}}$$

$$\bar{\mathbf{j}} \cdot \hat{\mathbf{n}} = 0$$

$$\bar{\mathbf{j}} = -\sigma \nabla \phi \quad \nabla \phi \cdot \hat{\mathbf{n}} = 0 \rightarrow \frac{\delta \phi}{\delta r} = 0$$



$$A \times = b$$

$$x \rightarrow \text{indexing?} \quad (N+1) \times (M+1)$$

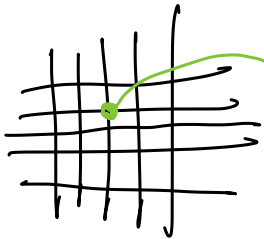
$$A \quad [(N+1)(M+1)], [(N+1)(M+1)]$$

$$x \quad [(N+1)(M+1)], 1$$

$$b \quad [(N+1)(M+1)], 1$$

$$x = \left[\underbrace{\phi_{1,1} \dots \phi_{1,M+1}}_{M+1} \underbrace{\phi_{2,1} \dots \phi_{2,M+1}}_{M+1} \dots \underbrace{\phi_{N+1,1} \dots \phi_{N+1,M+1}}_{M+1} \right]_{N+1}$$

h_1 (blue arrow to $\phi_{1,1}$)
 h (black arrow to $\phi_{2,1}$)
 h (blue arrow to $\phi_{N+1,M+1}$)



$$i, j \rightarrow h$$

$$h = (M+1)(i-1) + j$$

$$h = (M+1)(N) + M+1 = (M+1)(N+1)$$

$$i=1 \quad j=1$$

$$\phi_{1,1} \rightarrow h=1$$

$$i=N+1$$

$$j=M+1$$

$$\phi_{N+1,M+1}$$

$$i \neq 1 \longrightarrow h \neq (n+1)$$

$$j \neq 1 \longrightarrow h \neq 1$$